Case Study

Balancing Environmental and Economic Objectives for In-Stream Construction Projects: A Case Study on Sediment Management

Gregory Courtice[1] and Bernard Bauer[2]

Abstract: Balancing economic efficiency with ecological risk management is a key challenge of the civil construction industry, especially for activities within the wetted perimeter of rivers (*in-stream construction*). Suspended sediment (SS) releases may occur and be harmful to aquatic flora and fauna. A tradeoff exists when completing in-stream construction quickly (shorter duration of exposure, DoE; h) with a more intense release (higher SS concentration, SSC; $mg \cdot L^{-1}$) versus completing the same activity more slowly with a less intense release. Environmental regulations often prioritize lower SSC limits, sacrificing economic efficiency. However, there may be preferable alternatives that balance economic and environmental objectives. This paper presents a study conducted during the construction of a gravel cofferdam in the Bow River, Calgary, Canada that compared two different gravel placement methods: slow placement by an excavator bucket and a faster method using a bulldozer to push gravel in the river. An SS yield parameter was introduced to contrast earthworks productivity and sediment loading characteristics by normalizing SS dose [product of SSC and DoE, suspended sediment dose (SSD); $mg \cdot h \cdot L^{-1}$] to the quantity of gravel moved. The bulldozer method yielded greater economic efficiency without increased ecological risk associated with SS exposure, as defined by the management objective to protect salmonid fishes. This case study demonstrates the value of considering SS management tradeoffs by defining an operating limit using both SSC and DoE (or simply SSD) to achieve improved economic and environmental outcomes. However, implementing such an approach may face feasibility challenges depending on regulatory jurisdictions and may require changes in management policy to accommodate short-duration, high-intensity SS releases. **DOI: 10.1061/AOMJAH.AOENG-0033**. *This work is made available under the terms of the Creative Commons Attribution 4.0 International license, https://creativecommons.org/licenses/by/4.0/.*

This work is made available under the terms of the Creative Commons Attribution 4.0 International license.

Introduction

In the construction industry, time is money. Any factor or contingency that extends the project time-to-completion may lead to economic inefficiency and cost overruns. Thus, a primary objective of project management is to maximize satisfactory project outcomes by applying tactical decision-making that leads to streamlined operations (Pinto and Prescott 1990). Activity duration is a key parameter to manage because contractors have finite resources and are often faced with complex scheduling requirements (Kong and Dou 2021).

Construction projects must also satisfy a broad range of regulatory requirements, including those pertaining to the environment. Compliance with environmental regulations may trigger additional mitigation activities that can extend project timelines. These complexities are often considered in the time–cost tradeoff literature. Completing a project quickly may provide economic benefits, but it may also require a sacrifice in quality, safety risks to construction workers, or amplified environmental impacts.

This time–cost tradeoff is particularly relevant for in-stream construction projects conducted within the wetted perimeter of a river, such as bridge installation, riverbank stabilization, or habitat restoration. For these activities, environmental protection measures are often written into regulatory permits to preserve the integrity of sensitive aquatic ecosystems by managing suspended sediment (SS) releases (OECD 2007). Installation and removal of sediment barriers (Chapman et al. 2014; Madanayak and Sivakugan 2019) are often necessary to mitigate the impacts of heavy equipment operating in rivers and streams (Courtice and Naser 2020; Courtice et al. 2022a). This added scope of work extends project duration and hence affects economic efficiency.

Environmental regulations for in-stream projects usually require suspended sediment concentration (SSC) levels to be maintained below a prescribed threshold (CCME 2002; USEPA 2003). However, even when best practices are followed, exceedances of such SSC thresholds may still occur, requiring suspension of operations to allow for reductions in SSC levels below guidelines. Work stoppages are typically recommended by an on-site qualified environmental professional who makes judgment calls based on SSC sampling within the vicinity of the work site (Courtice and Naser 2020; Courtice et al. 2019). While this approach may satisfy environmental regulations, it often leads to economic inefficiencies. Thus, there is an inherent tension between environmental and economic objectives within in-stream construction project management.

Conceptual Background

The time–cost tradeoff problem has been studied extensively over the last 60 years in the construction industry using a variety of analytical methods (Mahdavian 2021; Bettemir and Birgönül 2017; Feng et al. 1997; Fulkerson 1961; Jiang and Zhu 2010; Kelley 1961; Mokhtari et al. 2011; Siemens 1971; Yang 2007). Completing construction activities quickly may come at the cost of quality,

[1]Applied Ecohydraulics, Kelowna V1Y 9Y9, Canada (corresponding author). ORCID: https://orcid.org/0000-0001-6880-4871. Email: greg.courtice@gmail.com

[2]Dept. of Earth, Environmental, and Geographic Sciences, Univ. of British Columbia, Kelowna V1V 1V7, Canada.

Note. This manuscript was submitted on November 30, 2023; approved on April 1, 2024; published online on June 6, 2024. Discussion period open until November 6, 2024; separate discussions must be submitted for individual papers. This paper is part of the ***ASCE OPEN: Multidisciplinary Journal of Civil Engineering***, ASCE, ISSN 2995-4266.

energy, or environmental effects (Lotfi et al. 2022). Multicriteria decision-making is part of the foundation of project planning to achieve ideal project outcomes. The ability to make effective, informed decisions in this framework is predicated on representative models for all project considerations, such as resource scheduling constraints, construction methods, health and safety systems, and environmental protection (Erdogan et al. 2019). Studies on this topic may be economically motivated by the potential to find optimal solutions that reduce activity timelines or minimize resources while achieving acceptable societal objectives. Yet, in the case of river construction, the environmental systems and sensitivities remain understudied and untested for these optimal frontiers in the construction management decision space.

A common regulatory requirement for in-stream construction projects is to maintain SSC below some threshold value, such as $25\ \mathrm{mg \cdot L^{-1}}$ over a 24-h period (ANZECC 2000; CCME 2002; USEPA 2012). Guidelines specifying point-exposure limits for SS were adopted by regulatory agencies after assessing the findings of multiple studies on aquatic ecosystems that documented negative effects due to varying levels of SS exposure (USEPA 2003). These observations include adverse responses by fish (e.g., behavioral, physiological, lethal), detrimental impacts on egg incubation, degradation of habitat quality, and turbidity effects on the intensity of sunlight for aquatic vegetation photosynthesis (Dudgeon et al. 2006; Kemp et al. 2011; Kjelland et al. 2015; Newcombe and Jensen 1996; Wood and Armitage 1997; Young and Mackie 1991).

Within the scientific literature, there is tacit recognition of the interdependency of SSC and duration of exposure (DoE) when assessing exposure–response dynamics of aquatic organisms (Courtice et al. 2022a, b; Newcombe and Jensen 1996; Suter 2007). Short-term exposure to high SSC may not be as harmful as long-term exposure to low SSC, depending on the precise SSC values and cumulative times of exposure. Nevertheless, in construction practice, often instantaneous or short-term measurements of SSC (e.g., once every hour while the exposure is present) are used to determine whether a regulatory guideline has been exceeded, which then may trigger the need for additional mitigation measures or temporary work stoppage. The potential tradeoffs between SSC and DoE have yet to be implemented as a viable option for project management, largely because the dynamics of exposure–response relationships are poorly understood. Thus, the status quo prevails.

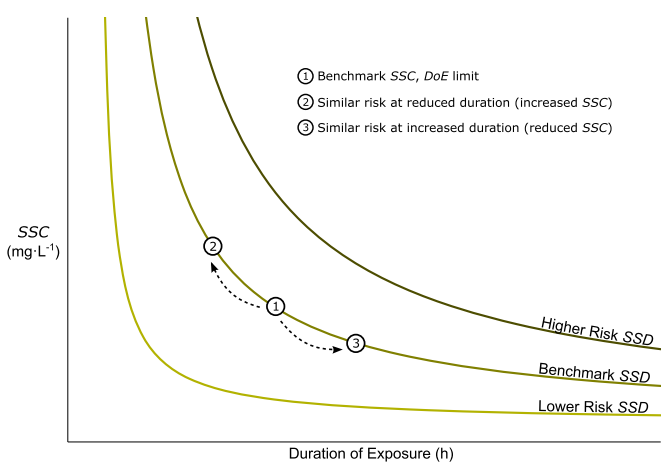

Fig. 1. Conceptual management operating space using SSC versus DoE to describe the relative risk of adverse effects to fish based on findings of Courtice et al. (2022a, b). The benchmark SSD is prescribed by regulatory guidelines. Three hypothetical points on the benchmark contour represent: (1) the regulatory guideline for a point-exposure limit, such as $25\ \mathrm{mg \cdot L^{-1}}$ over a 24 h duration; (2) a point exposure with similar risk as the benchmark SSD but with increased SSC over a shorter project duration; and (3) a point exposure with similar risk but with decreased SSC over a longer project duration.

Recently, Courtice et al. (2022a, b) proposed a suspended sediment dose (SSD) model that explicitly considered the tradeoff between SSC and DoE when managing SS exposure risk to aquatic fauna. The basis for the model is that any prescribed regulatory benchmark, such as $25\ \mathrm{mg \cdot L^{-1}}$ over a 24-h period, can be expressed as an equivalent SSD (i.e., $600\ \mathrm{mg \cdot h \cdot L^{-1}}$), which can, in turn, be represented as a single contour line on a graph of SSC versus DoE (Fig. 1). Along a single SSD contour, any SSC–DoE combination that satisfies the SSD condition will pose a similar risk of adverse effects in organism response. Courtice et al. (2022a, b) suggested that SSD provides a more representative and operationally manageable compliance limit for the risk of adverse effects to salmonids than those focused on SSC alone. In addition, the flexibility to manage across a range of SSC and DoE values may have environmental benefits while allowing project managers to achieve better economic efficiencies. These assertions have yet to be tested, which is the purpose of this case study.

The objective of this paper is to present the first practical test of the assertion that using SSD to inform management decisions—rather than strictly relying on an SSC-based regulatory guideline—can provide opportunities to achieve desirable environmental and economic project outcomes simultaneously. The paper is organized into five sections, as follows: (1) Project Description; (2) Methodology and Metrics; (3) Results; (4) Discussion; and (5) Conclusions.

Project Description

The City of Calgary, Alberta, Canada proposed to construct earthworks within the wetted perimeter of the Bow River as part of an aquatic habitat enhancement project (Fig. 2). A sediment barrier (gravel cofferdam) was to be constructed using local gravel from an adjacent bar within the channel. Placement of unwashed river gravel by heavy machinery within the wetted perimeter was anticipated to release a large quantity of fine sediment, whereas prewashing the gravel was deemed cost-prohibitive. There was concern that releasing large quantities of fine sediment would exceed

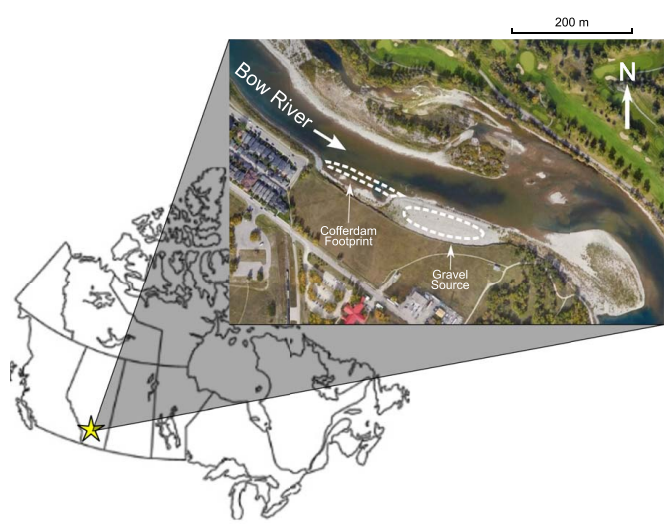

Fig. 2. Study location along the Bow River in Calgary, Alberta, Canada. (Base image © Google.)

This work is made available under the terms of the Creative Commons Attribution 4.0 International license.

typical regulatory limits on SSC and thereby increase the risk of adverse effects on the Bow River ecosystem. Substantial cost overruns were anticipated if the construction contractor was required to remain compliant with the regulatory SSC limit by restricting the rate of gravel placement.

The SS compliance framework common to this management jurisdiction (Alberta Environment and Protected Areas—South Saskatchewan Region) is based on an interpretation of recommendations provided by the Canadian Council of Ministers of the Environment (CCME) water quality guidelines for the protection of aquatic life (Alberta Queen's Printer 2019; CCME 2002). Under this framework, construction activities are halted upon measuring SSC values above 25 $mg{\cdot}L^{-1}$ until the limit is no longer exceeded. This limit is often applied to SSC above background values where natural SS is present. Any occurrence of an exceedance must also be called into the environmental response hotline, formally documented, and reported to the provincial regulator with additional mitigation plans to avoid future exceedances.

After considering alternatives, the City of Calgary adopted a suspended sediment management framework based on Courtice and Naser (2020) and Courtice et al. (2019) that focused on daily duration limits rather than the standard SSC-based regulatory threshold approach. The underlying basis for this approach is that SS exposure–response dynamics in fish are governed by both SSC and DoE, as illustrated in Fig. 1. This conceptual framework was translated into a management strategy by prescribing daily duration limits for operations based on time-weighted mean SSC values to ensure that the construction activities would not exceed a risk tolerance objective. The risk tolerance was informed by CCME (2002) guidelines and assessed site-specific ecosystem sensitivities by the project's environmental consultant (Matrix, "Water Quality Monitoring Plan for Use during the Bow River Restricted Activity Periods, Inglewood Bird Sanctuary Reconnection Project," internal memo submitted to City of Calgary, March 2022.). The Newcombe and Jensen (1996) *Severity of Ill-Effects* scale and multiple linear regression models were used to help establish management objectives for the protection of aquatic life consistent with the judgment of the environmental consultant. A corresponding duration-based operating limit was then specified based on the management objectives. A graphical illustration of this management approach is presented in Fig. 3, which includes corresponding SSC–DoE values derived for regulatory compliance purposes.

By setting DoE operating limits that vary as a function of SSC, construction management decisions can be made to achieve desirable economic and environmental objectives simultaneously. Moreover, the flexibility inherent to duration-based management provides the opportunity to investigate the implications of potential tradeoffs between SSC and DoE, especially regarding maximizing project efficiency (i.e., reducing DoE by increasing allowable SSC).

Methods and Metrics

The gravel cofferdam was constructed over 3 days using two different construction methodologies representative of (1) a conventional strategy to reduce plume intensity by slowing the introduction rate of gravel during cofferdam construction (Day 1, April 4, 2022) and (2) a strategy following duration-based management informed by the SS exposure–response modeling of Courtice et al. (2022a, b) to achieve both environmental and economic objectives (Day 2 and Day 3, April 5 and 6, 2022). Only the first 2 days of construction are considered here because the site conditions were similar (4–5 m flow depth with similar hydraulic characteristics) on these 2 days (April 4 and 5, 2022). Construction activities on Day 3 (April 6, 2022) were located in shallower flow conditions with different hydraulic characteristics; therefore, a direct comparison is not appropriate.

Construction Methods

Day 1 of the cofferdam construction prioritized the reduction of SS plume intensity. Dump trucks supplied gravel on dry ground next to an excavator, which was located on the edge of the wetted

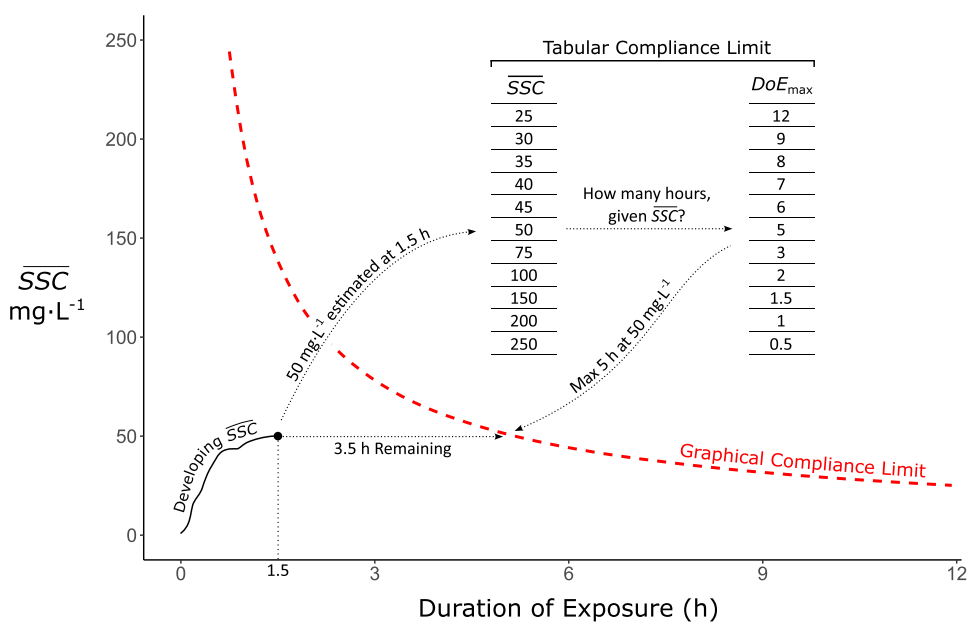

Fig. 3. Graphical representation of the duration-based SS management framework using the case study-specific compliance limit. A hypothetical exposure over an initial work period is presented with its developing $\overline{SSC}$ time series. At each point in time of interest (e.g., 1.5 h), an estimated $\overline{SSC}$ is calculated given all SSC observations recorded up to the time of interest. Second, a corresponding maximum daily duration of exposure (DoE_{max}) is identified that corresponds to the $\overline{SSC}$ value. Finally, DoE_{max} is subtracted by the current DoE to determine the remaining duration.

This work is made available under the terms of the Creative Commons Attribution 4.0 International license.

perimeter and was used to place gravel in the river to build the cofferdam. Working slowly and methodically, this method precisely controlled the placement rate and location of gravel introduction in the wetted perimeter.

Day 2 of the cofferdam construction emphasized project efficiency over SS plume intensity. A bulldozer was used to push large quantities of gravel into the river rather than the more careful and gradual placement of small quantities by the excavator. The gravel was also supplied by truck, but more frequent and numerous loads were required due to the large volume of material able to be moved by the bulldozer. Several individuals on site had expressed concerns prior to implementation regarding potential negative project outcomes with respect to this faster placement method. The concern was primarily related to the possibility of extreme SSC values that would yield very short daily duration limits (as per the duration-based management strategy). This very short daily duration limit would then hypothetically compromise the ability to complete the project in a timely fashion.

Data Collection

Two primary data streams were collected during the project: (1) the quantity and rate of gravel placement throughout the day; and (2) the SSC time series at sampling sites close to the construction area. In situ turbidity samples were collected to quantify SSC using a site-specific SSC-NTU rating curve developed prior to construction. Gravel placement was tracked using construction activity notes recorded by the environmental consultant.

In Situ Turbidity Sampling

Water turbidity (NTU) was sampled every 15 min upstream and downstream of the activity site using In Situ Aqua TROLL 500 water quality sondes. Background samples were taken immediately upstream of the site, while downstream samples were taken approximately five river-widths (~500 m) downstream. This downstream distance was selected as representative of an acceptable mixing zone, given river characteristics and ecosystem sensitivities at the time of construction (Matrix, "Water Quality Monitoring Plan," 2022). Measurement error on the turbidity sondes is reported as the greater of $\pm 2\%$ or ± 0.5 NTU; however, due to the variability in particle densities in a natural environment, a more conservative measurement error of $\pm 5\%$ was applied to the study analysis. SSC was calculated from NTU based on a site-specific calibration curve developed by on-site personnel.

Quantity of Gravel Deposition Time Series

Daily quantities of gravel placed in the river were used to estimate a cumulative quantity time series by discretizing the total daily volume recorded by the construction contractor into equivalent 15-min intervals to coincide with turbidity sampling intervals. For periods of inactivity, as documented in the consultant's notes, the time series was recorded as having no increased volume of deposited gravel. These notes may not coincide precisely with construction activities; however, they provide a reasonable representation of the management decisions made throughout the day.

Duration of Exposure Estimation

Daily DoE was calculated by estimating the number of hours (to the nearest 15-min interval) that elevated levels of SSC were present in the water column at the turbidity measurement site. Upon completion of daily construction activities, the SS plume was deemed to have passed through the system once SSC values returned to within $5\ \mathrm{mg \cdot L^{-1}}$ of background levels. This $5\ \mathrm{mg \cdot L^{-1}}$ threshold was taken to be consistent with measurement error at the low range of turbidity measurements.

River Discharge

Bow River discharges for the 3-day study period were not measured but were available from government-operated hydrometric stations. The data were downloaded and reviewed to verify that SSC dilution, or other impacts to sediment entrainment characteristics, did not occur from temporal variation in discharge. Water Survey of Canada stations 05BB001 (Bow River at Calgary) and 05BJ004 (Elbow River below the Glenmore Dam) were summed to obtain an estimated discharge hydrograph at the study site, located approximately 3.5 km downstream of the Bow–Elbow confluence. No major discharge contributions are present in the system between the published hydrometric data and the study site.

Analytical Parameters

Mean Suspended Sediment Concentration

A time-weighted mean SSC ($\overline{\mathrm{SSC}}$) was estimated using SSC observations derived from turbidity samples at 15-min intervals. A time series for $\overline{\mathrm{SSC}}$ was then graphed to assess the downstream loading during the day relative to regulatory compliance limits and ecological risk tolerances based on recommended guidelines (CCME 2002) and SS exposure–response models for salmonids (Courtice et al. 2022a, b).

Suspended Sediment Dose

An SS dose parameter (SSD) was calculated as the product of $\overline{\mathrm{SSC}}$ and DoE. This parameter was used to assess the relative mass of sediment introduced into the flow for a given SS release day. Based on the principle of mass conservation, changes in SSD are proportional to the quantity of fine sediment released from the project site. Key assumptions for comparing one day to another include steady discharge conditions and a fixed distance between the release and the monitoring locations. The SSD parameter was identified by Courtice et al. (2022b) as the most representative of the risk of adverse effects on salmonids from SS, relative to SSC or DoE, separately.

Suspended Sediment Yield

An SS yield parameter (Y_{SS}) was calculated by dividing SSD by the daily quantity of gravel placement to assess the potential for SS to be released per unit of gravel introduced into the river, given that there were differences in the placement method. The null hypothesis would be that, all else being equal, Y_{SS} should be the same regardless of placement method and the total quantity of gravel placed in the river. A smaller Y_{SS} value for one method over another would suggest a more desirable approach that reduces SS loading when accounting for the total quantity of deposited gravel.

This work is made available under the terms of the Creative Commons Attribution 4.0 International license.

Results

The scope of construction and SS plume characteristics that occurred over the 2-day construction period are shown in Fig. 4. Day 1 used a strategy that prioritized a reduction in SS plume intensity by carefully placing gravel with an excavator bucket. This strategy resulted in a longer DoE with a lower $\overline{SSC}$ and smaller quantity of deposited gravel. Day 2 used a bulldozer and resulted in increased plume intensity with a higher $\overline{SSC}$ and shorter DoE, but a much larger quantity of gravel was placed.

Fig. 5 presents the discharge hydrograph for the study site based on the cumulative discharge of the Bow River and Elbow River monitoring stations. A $\pm 5\%$ Q_{mean} band is presented with dotted lines on the time series, indicating that the discharge remained within this range for most of the study period. Thus, the effects of dilution on SSC measurements were unlikely to occur due to varying discharge or altered mixing effects.

Fig. 6 presents time series graphs for the cumulative volume of gravel placement (upper), SSC observations estimated from in situ turbidity samples (middle), and $\overline{SSC}$ calculated from SSC observations (lower). Construction activities on Day 1 (April 4, 2022) had two forced work stoppages to manage environmental risk when SSC plume intensity was high. On Day 2 (April 5, 2022), construction activities were continuous with no stoppages until the planned work was completed. Both days achieved regulatory compliance, but nearly three times as much gravel was placed on Day 2 as on Day 1. Moreover, since construction was not halted on Day 2, the duration of in-stream activity was reduced by a factor of two, which leads to project efficiency and provides a longer recovery period for the river to attain and hold background turbidity values.

Suspended sediment yield values, Y_{SS}, were estimated to be 0.44 $\pm 0.02\ \text{mg} \cdot \text{h} \cdot \text{L}^{-1} \cdot \text{m}^{-3}$ (Day 1) and $0.12 \pm 0.01\ \text{mg} \cdot \text{h} \cdot \text{L}^{-1} \cdot \text{m}^{-3}$ (Day 2). The difference in Y_{SS} values suggests that the excavator method used on Day 1 generated more SS per volume of gravel

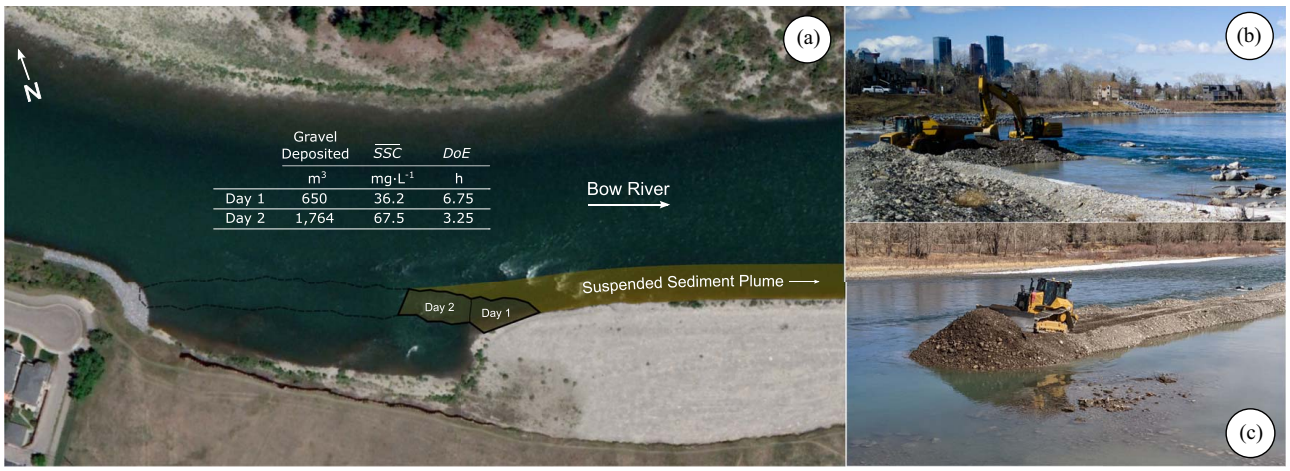

Fig. 4. (a) Graphic summary of suspended sediment plume characteristics (inset table) with photos of construction from (b) Day 1 (excavator); and (c) Day 2 (bulldozer). The cofferdam footprint constructed on a third day (not included in the study) is presented with a dashed line on part (a). The inset table shows the volume of gravel deposited each day, $\overline{SSC}$, and DoE. [Base image (a) © Google; images (b, c) by Gregory Courtice.]

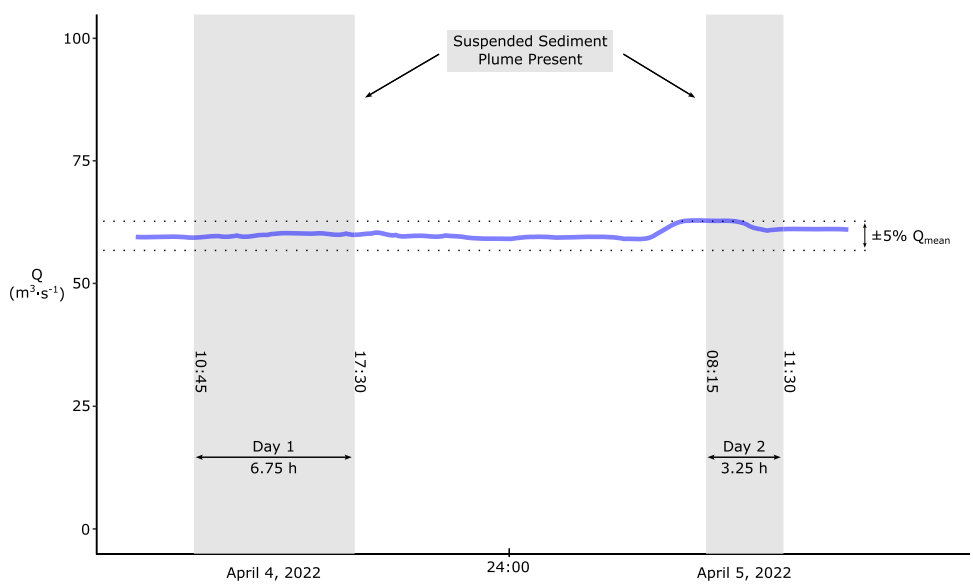

Fig. 5. Discharge hydrograph at the study site estimated from Water Survey of Canada monitoring stations on the Bow River at Calgary (05BB001) and the Elbow River below the Glenmore Dam (05BJ004). Periods where a suspended sediment plume was present downstream of construction are identified as gray-shaded regions. The dotted lines represent discharge values that are $\pm 5\%$ of the mean discharge (Q_{mean}) for the study period.

This work is made available under the terms of the Creative Commons Attribution 4.0 International license.

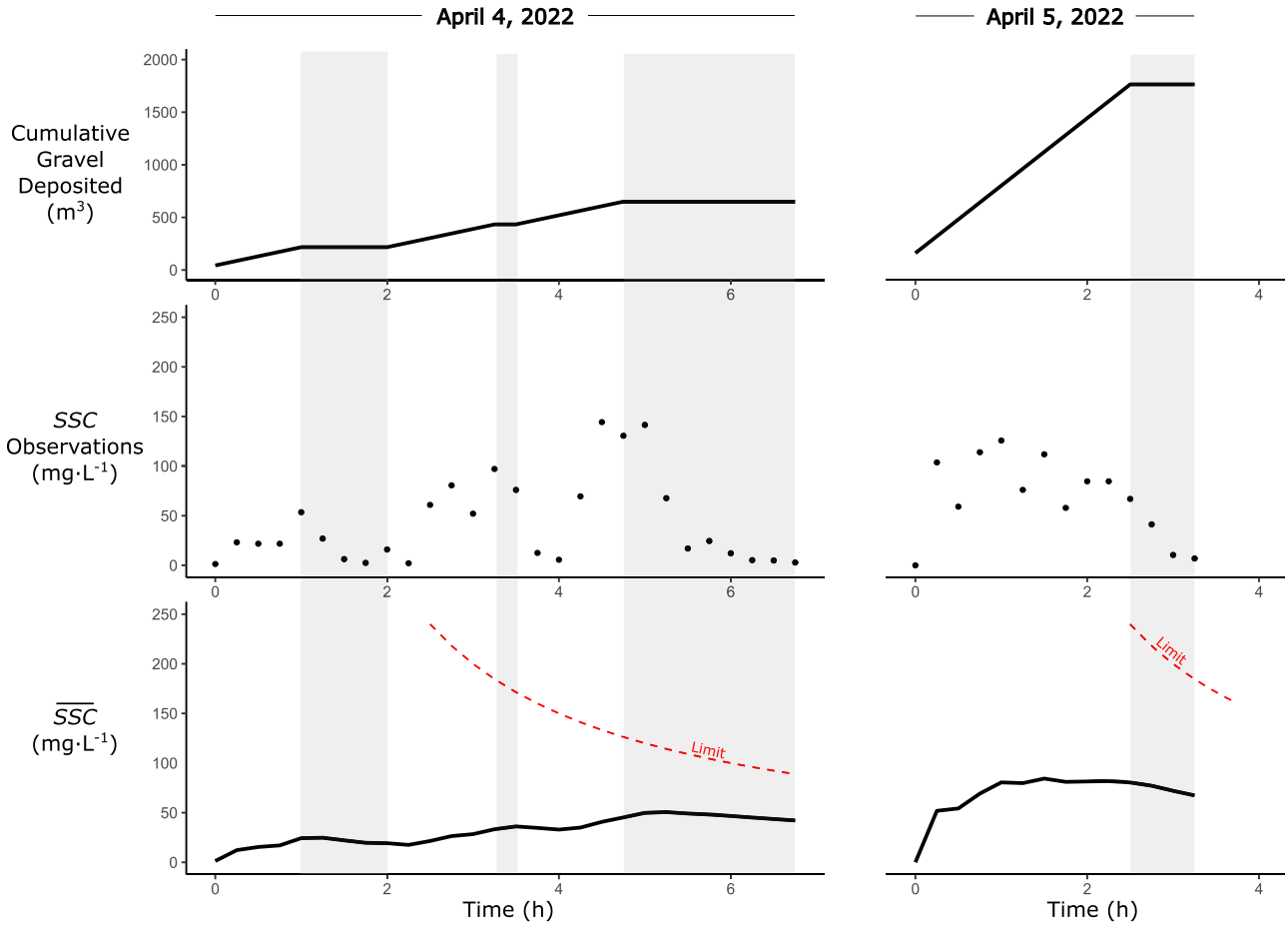

Fig. 6. Time series of the cumulative volume of gravel deposition, SSC observations, and $\overline{SSC}$ calculated from SSC observations. Gray-shaded regions correspond to periods when construction was halted for management purposes to reduce SSC. Dashed lines on the $\overline{SSC}$ time series represent the duration-based compliance limit established for the project, which decreases as the duration of exposure increases.

placement than the bulldozer method used on Day 2. The smaller value on Day 2 suggests that fine sediment remains trapped within the placement volume and does not become entrained into the flow.

Discussion

The $\overline{SSC}$-DoE exposure space, based on the conceptual diagram presented in Fig. 1, is presented (Fig. 7) to highlight the tradeoff between environmental and economic objectives for the two gravel placement methods. A time series of $\overline{SSC}$ values is shown for Day 1 and Day 2 as construction activities progressed through each day, with circles plotted on the coordinates of the final $\overline{SSC}$-DoE values. The cumulative gravel placement is also included as a ribbon graphed above the corresponding $\overline{SSC}$ time series, with the height of the ribbon at the end of the construction day in proportion to the total volume of gravel placement for that day. The SSD contour (red dotted line) corresponds to the Day 2 (April 5, 2022) end-of-day value as a means to visually compare the release characteristics for each day.

Fig. 7 represents a concise graphical summary that can be used to answer a series of questions regarding the tradeoff between economic and environmental objectives, as follows:

1. Will improving construction efficiency increase the number of earthworks completed each day, or does the corresponding decrease in daily duration limit due to increased SS release intensity negate efficiency benefits?

   The gray ribbon above the Day 2 time series is nearly triple the height of the ribbon above Day 1, illustrating that a larger quantity of gravel was placed on Day 2 at less than half the time. Project efficiency was therefore better with the bulldozer method.
2. Does the method of earthworks construction influence the mass of SS released when accounting for the gravel volume placed?

   The final exposure on Day 1 is above the 250 $mg \cdot h \cdot L^{-1}$ reference SSD contour while the Day 2 exposure is below this same reference contour. Given that the volume of gravel placement was much less on Day 1, it appears that using an excavator to place gravel results in greater SS loading than the bulldozer method when the total volume of gravel is taken into consideration.
3. Which method of gravel placement is likely to have a reduced risk of adverse effects on salmonids, given the SS exposure–response models developed by Courtice et al. (2022a, b)?

   The end-of-day SSD value for Day 1 is larger than Day 2, which suggests a greater risk of adverse effects according to the SSD-based modeling of Courtice et al. (2022a, b). Further, the Day 2 approach provides longer rest periods between subsequent activity days due to the shorter DoE.

Simplifying SS Management with a Suspended Sediment Dose Limit

In-stream construction activities following traditionally applied SS management practices must adhere to regulatory requirements that

This work is made available under the terms of the Creative Commons Attribution 4.0 International license.

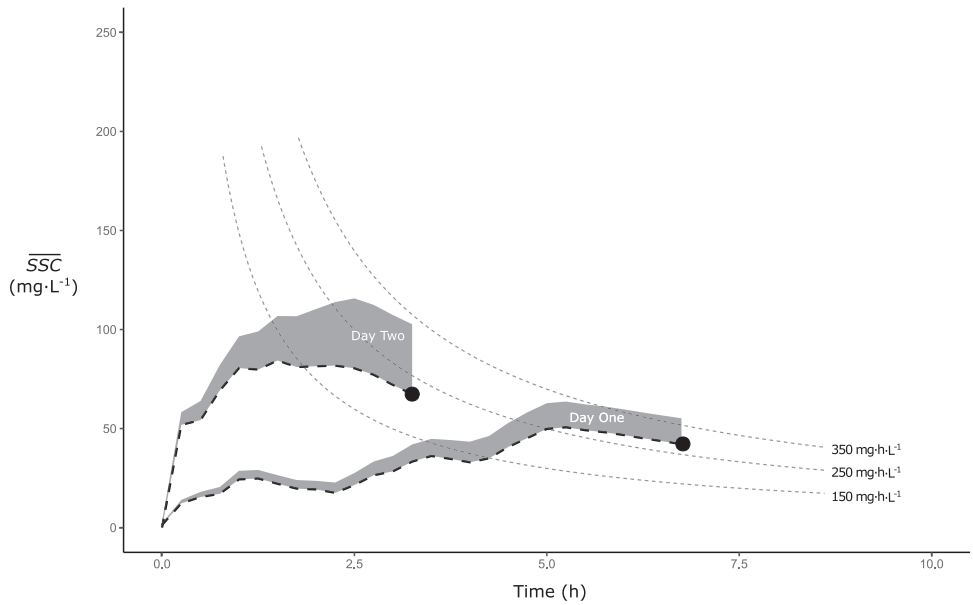

Fig. 7. Graphical summary of results. $\overline{SSC}$ time series are graphed for each day using a dashed line. The gray-shaded area above the $\overline{SSC}$ time series represents cumulative gravel placement during the construction activity. Reference SSD contours (dotted lines) are included to compare SS loading between the two construction days, given that SSD can be used as a surrogate parameter for sediment loading when mixing conditions and monitoring locations are held constant. Day 1 falls above the 250 mg·h·L^{-1} contour, while Day 2 falls below the same contour, indicating Day 1 experienced a larger SS loading event than Day 2.

prioritize maintaining SSC values below a threshold level. This approach may be overly constrained, as suggested by Courtice et al. (2022a, b) and supported by this study. An SSD limit, such as 600 mg·h·L^{-1} (corresponding to 25 mg/L over 24 h; CCME 2002), can be adapted to provide greater management flexibility without a likely increase in risk to environmental outcomes (in this case, specifically for salmonid SS exposure–response dynamics). Guidance to SS management derives from selecting an acceptable SSD limit based on the level of risk aversion and dividing SSD by $\overline{SSC}$ to obtain a target maximum DoE for the day's activities (DoE_{max}). There is, however, a greater need for scientific studies to ensure that appropriate risk tolerances for the protection of aquatic life are understood across a range of SSD values and species. This can be accomplished by evaluating exposure–response models for specific species of interest within particular management regions using methods similar to those developed by Courtice et al. (2022a, b).

Summary and Conclusions

A gravel earthworks construction project in the Bow River in Calgary, Alberta, Canada provided the opportunity to investigate whether the tradeoffs between economic and environmental objectives revealed by the modeling of Courtice et al. (2022a, b) can be realized in practice. Two construction methods are used to construct a cofferdam to manage SS releases from a project intended to enhance fish habitat. The first day employed a common method of managing SSC (mg·L^{-1}) by using an excavator to reduce the intensity of gravel placed in the river. The next day employed a bulldozer to push large volumes of gravel into the river, which led to a reduction in the duration of exposure (DoE, h), thereby increasing project efficiency. The study findings suggest that the bulldozer achieved better overall outcomes, including larger volumes of gravel placement over a shorter period and a lower yield of suspended sediment introduced into the river. Moreover, the final SSD (mg·h·L^{-1}) values are lower, indicating a likely reduced risk of negative environmental consequences. Implementing such an approach requires a relaxation of conventional regulatory SSC limits to accommodate short-duration, high-intensity SS releases. Many jurisdictions may be reluctant to allow for such regulatory relaxations until more examples of similar case studies with desirable environmental outcomes have been documented and shared broadly. This case study provides supporting evidence to researchers and practitioners for the relaxation of traditional SSC or turbidity-based limits to investigate management strategies that more optimally achieve both environmental and economic project objectives.

Data Availability Statement

Some or all data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

The authors thank the City of Calgary for their willingness to apply innovative environmental practices to the case study project, in addition to providing funding that assisted with the development of this manuscript.

References

Alberta Queen's Printer. 2019. "Code of practice for watercourse crossings." In *Water act and the water (ministerial regulation)*. Edmonton, AB: Alberta Queen's Printer.

ANZECC (Australian and New Zealand Environment and Conservation Council). 2000. *National water quality management strategy,*

This work is made available under the terms of the Creative Commons Attribution 4.0 International license.

Australian and New Zealand guidelines for fresh and marine water quality. Canberra, Australia: ANZECC.

Bettemir, Ö. H., and M. T. Birgönül. 2017. "Network analysis algorithm for the solution of discrete time-cost trade-off problem." *KSCE J. Civ. Eng.* 21 (4): 1047–1058. https://doi.org/10.1007/s12205-016-1615-x.

CCME (Canadian Council of Ministers of the Environment). 2002. *Canadian water quality guidelines for the protection of aquatic life*. Winnipeg, MB: CCME.

Chapman, J. M., C. L. Proulx, M. A. N. Veilleux, C. Levert, S. Bliss, M.-E. Andre, N. W. R. Lapointe, and S. J. Cooke. 2014. "Clear as mud: A meta-analysis on the effects of sedimentation on freshwater fish and the effectiveness of sediment-control measures." *Water Res.* 56: 190–202. https://doi.org/10.1016/j.watres.2014.02.047.

Courtice, G., B. Bauer, C. Cahill, B. Naser, and A. Paul. 2022a. "Suspended sediment releases in rivers: Toward establishing a safe sediment dose for construction projects." *Sci. Total Environ.* 848: 157685. https://doi.org/10.1016/j.scitotenv.2022.157685.

Courtice, G., B. Bauer, C. Cahill, B. Naser, and A. Paul. 2022b. "A categorical assessment of dose-response dynamics for managing suspended sediment effects on salmonids." *Sci. Total Environ.* 807 (1). https://doi.org/10.1016/j.scitotenv.2021.150844.

Courtice, G., B. Bauer, G. Naser, and D. Roberts. 2019. "Development of a duration-based management framework for in-stream construction-induced suspended sediment." In *Proc., 38th Int. Association for Hydro-Environment Engineering and Research World Congress*. Panama City, Panama: International Association for Hydro-Environment Engineering and Research.

Courtice, G., and G. Naser. 2020. "In-stream construction-induced suspended sediment in riverine ecosystems." *River Res. Appl.* 36: 327–337. https://doi.org/10.1002/rra.3559.

Dudgeon, D., et al. 2006. "Freshwater biodiversity: Importance, threats, status and conservation challenges." *Biol. Rev.* 81 (2): 163–182. https://doi.org/10.1017/S1464793105006950.

Erdogan, S. A., J. Šaparauskas, and Z. Turskis 2019. "A multi-criteria decision-making model to choose the best option for sustainable construction management." *Sustainability* 11 (8): 2239. https://doi.org/10.3390/su11082239.

Feng, C.-W., L. Liu, and S. A. Burns. 1997. "Using genetic algorithms to solve construction time-cost trade-off problems." *J. Comput. Civil Eng.* 11 (3): 184–189. https://doi.org/10.1061/(ASCE)0887-3801(1997)11:3(184).

Fulkerson, D. R. 1961. "A network flow computation for project cost curves." *Manage. Sci.* 7 (2): 167–178. https://doi.org/10.1287/mnsc.7.2.167.

Jiang, A., and Y. Zhu. 2010. "A multi-stage approach to time-cost trade-off analysis using mathematical programming." *Int. J. Construct. Manage.* 10 (3): 13–27. https://doi.org/10.1080/15623599.2010.10773147.

Kelley, J. E. 1961. "Critical-path planning and scheduling: Mathematical basis." *Oper. Res.* 9 (3): 296–320. https://doi.org/10.1287/opre.9.3.296.

Kemp, P., D. Sear, A. Collins, P. Naden, and I. Jones. 2011. "The impacts of fine sediment on riverine fish." *Hydrol. Processes* 25: 1800–1821. https://doi.org/10.1002/hyp.7940.

Kjelland, M. E., C. M. Woodley, T. M. Swannack, and D. L. Smith. 2015. "A review of the potential effects of suspended sediment on fishes: Potential dredging-related physiological, behavioral, and transgenerational implications." *Environ. Syst. Dec.* 35: 334–350. https://doi.org/10.1007/s10669-015-9557-2.

Kong, F., and D. Dou. 2021. "Resource-constrained project scheduling problem under multiple time constraints." *J. Constr. Eng. Manage.* 147 (2). https://doi.org/10.1061/(ASCE)CO.1943-7862.0001990.

Lotfi, R., Z. Yadegari, S. Hosseini, A. Khameneh, E. Tirkolaee, and G. W. Weber. 2022. "A robust time-cost-quality-energy-environment trade-off with resource-constrained in project management: A case study for a bridge construction project." *J. Ind. Manag. Optim.* 18 (1): 375–396. https://doi.org/10.3934/jimo.2020158.

Madanayak, T. A., and N. Sivakugan. 2019. "Simple solutions for square and rectangular cofferdam seepage problems." *Can. Geotech. J.* 56 (5): 730–745.

Mahdavian, S. 2021. "Time-Cost Trade-off (TCT) analysis based application framework for safety centric construction acceleration planning." Master of Science thesis. Dept. of Civil and Environmental Engineering, Univ. of Alberta.

Mokhtari, H., R. B. Kazemzadeh, and A. Salmasnia. 2011. "Time-cost tradeoff analysis in project management: An ant system approach." *IEEE Trans. Eng. Manage.* 58 (1): 36–43. https://doi.org/10.1109/TEM.2010.2058859.

Newcombe, C. P., and J. O. T. Jensen. 1996. "Channel suspended sediment and fisheries: A synthesis for quantitative assessment of risk and impact." *North Am. J. Fish. Manage.* 16: 693–727. https://doi.org/10.1577/1548-8675(1996)016<0693:CSSAFA>2.3.CO;2.

OECD (Organization for Economic Cooperation and Development). 2007. "Guiding principles of effective environmental permitting systems." *Organization for Economic Cooperation and Development*. Accessed April 6, 2023. https://www.oecd.org/env/outreach/37311624.pdf.

Pinto, J. K., and J. E. Prescott. 1990. "Planning and tactical factors in the project implementation process." *J. Manage. Stud.* 27: 305. https://doi.org/10.1111/j.1467-6486.1990.tb00249.x.

Siemens, N. 1971. "A simple CPM time-cost tradeoff algorithm." *Manage. Sci.* 17 (6): B-354–B-363. https://doi.org/10.1287/mnsc.17.6.B354.

Suter, G. W. 2007. *Ecological risk assessment*. 2nd ed. Boca Raton, FL: CRC Press.

USEPA. 2003. *Developing water quality criteria for suspended and bedded sediments (sabs)*. Washington, DC: USEPA Office of Water.

USEPA. 2012. "Water quality criteria." In *Water quality standards handbook*, 19–24. Washington, DC: USEPA.

Wood, P. J., and P. D. Armitage. 1997. "Biological effects of fine sediment in the lotic environment." *Environ. Manage.* 21 (2): 203–217. https://doi.org/10.1007/s002679900019.

Yang, I.-T. 2007. "Using elitist particle swarm optimization to facilitate bicriterion time-cost trade-off analysis." *J. Constr. Eng. Manage.* 133 (7): 498–505. https://doi.org/10.1061/(ASCE)0733-9364(2007)133:7(498).

Young, R. J., and G. L. Mackie. 1991. "Effect of oil pipeline construction on the benthic invertebrate community structure of Hodgson Creek, Northwest Territories." *Can. J. Zool.* 69: 2154–2160. https://doi.org/10.1139/z91-301.

This work is made available under the terms of the Creative Commons Attribution 4.0 International license.